

Chase Sun

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EDUCATION

Master of Science in Mechanical Engineering Georgia Institute of Technology, Atlanta, GA Anticipated graduation date: July 2026	August 2024 – Present
Bachelor of Engineering, Mechanical Engineering Carleton University, Ottawa, ON, Canada	September 2017 – May 2022

SKILLS

Programming: Embedded systems programming, C, C++, Assembly, MATLAB, Linux (debian)
CAD / CAE: SolidWorks, CATIA V5, AutoCAD, Fusion, Onshape, EasyEDA PCB design, ANSYS Workbench

PROFESSIONAL EXPERIENCE

National Research Council of Canada – Aerosol & Gas Metrology Lab Technical Officer Research Assistant	May 2022 – July 2024 January 2021 – August 2021
• Embedded development: Programmed with the Beagle Bone Black (BBB) single board computer in Linux to create high precision data capture for a lightweight, UAV mounted laser absorption spectroscopy methane sensor • Real time programming: Developed C and Assembly code on the BBB real-time processor to multithread simultaneous capturing and synchronization of high frequency optical measurement, ambient conditions, and GPS information through direct register interfacing and bit-banged SPI & I2C communication • Circuit design: Designed and assembled custom analog and digital printed circuit boards for signal conditioning, transimpedance amplification, precision control, and battery management • Field work experience: Designed and prototyped the physical sensor package and deployed sensors across Canada including the west arctic to map greenhouse gas emissions in collaboration with the Geological Survey of Canada and Environment and Climate Change Canada • Key results: Work presented at conferences in US and Canada and a second author publication is under review	

ProSlide Technology

Product Development Intern	May – December 2020
• CFD: Simulated water flow through flumes of commercial water slides using ANSYS Fluent • CAD: Created engineering drawings of water slide components using SolidWorks	

RESEARCH & TEACHING EXPERIENCE

Exoskeleton & Prosthetic Intelligent Control Lab – Georgia Tech Master's Thesis Student	August 2024 - present
• Researching and developing a cable-driven soft exoskeleton to protect and assist manual labor workers in hazardous environments • Sensor fusion: Programmed in C using a Teensy 4.0 microcontroller to interface with robotic actuators, inertial measurement units, and strain gauge amplifiers using multithreading for concurrent data capture and command • Controller design: Developed code for a finite state machine lifting controller coupled with a low-level admittance controller for assistance in lifting tasks • Soft textile prototyping: Prototyped the device using soft textiles and custom mounting interfaces • Key results: Conducted experiments on exoskeleton effectiveness in assisting lifting tasks and achieved ~30% reduction in EMG activation on the quadriceps muscles with a current sample size of 6 to be expanded	
Georgia Tech Woodruff School of Mechanical Engineering Lead Course Instructor Graduate Teaching Assistant	January 2026 - Present August 2024 – December 2025
• Lesson planning: Prepared and delivered lectures to classes of 250 senior ME students on topics relating to low-level controls, internal combustion engines, and refrigeration systems for the mechanical systems lab course • Technical Communication: Supervised labs to guide students working with real engineering systems and facilitated systems analysis, trouble shooting, and reporting at a professional level	

- **Mentorship:** Trained and managed groups of 8 teaching assistants to improve instructional quality
- First MS student in department history to become an official lead instructor

The Hive Electronic Makerspace at Georgia Tech

Embedded Systems Master Instructor

January 2025 - Present

- Taught and mentored students from diverse engineering backgrounds on embedded systems programming and mechatronics at the largest electronic student makerspace in the world
- Organized and delivered interactive embedded systems related workshops on C++, PCB design, serial communication, electromechanical systems, and Bluetooth low energy communications

PUBLICATIONS AND PRESENTATIONS

C. Sun, *"Teaching for Belonging: How TAs are the most important piece of the student engagement puzzle."* Keynote speaker, Georgia Tech TA and future faculty awards ceremony. Atlanta, GA, March 2026

M. Beattie, C. Sun, R. MacLeod, N. Sabourin, P. Morse, J. Corbin, J. Oliaee, *"Ultra-Lightweight Mid-IR Methane Sensor for UAV-based measurements"*, Journal of Optical Express, Submitted for review

J. Oliaee, R. MacLeod, C. Sun, M. Beattie, S. Dallimore, P. Morse, *"Measurement of natural methane emissions in the western Canadian arctic using an ultra-lightweight laser spectroscopic sensor aboard a small UAV"*, CabCH4 symposium, Ottawa, Canada, May 2024

C. Sun, J. Oliaee, N. Sabourin, G. Smallwood, *"Development of a Lightweight UAV-Mountable Open-Cell Methane Sensor Using Mid-Infrared Tunable Diode Laser Interfaced with a Low-Cost Embedded Platform."* Onboard Sensing, Analysis, and Reporting (OSAR) conference. Riverside, CA, March 2023

APPLIED PROJECTS

Blended Wing UAV Capstone Project, Carleton University

Team leader

September 2021 - April 2022

- Led a team of 22 students to design, manufacture, and test a blended wing UAV with vertical takeoff and landing (VTOL) capabilities
- **CAD & FEA:** Created CAD parts of blended wing UAV with CATIA V5 and collaborated with teams conducting stress analysis, CFD simulations, and 3D printing with the 3D model.
- **Project management:** Organized and moderated weekly design reviews, team meetings, and progress report presentations to lead engineers and industry professionals

Carleton University Design League - Sustainable Design Wall-e Challenge

Team Leader

March 2022

- Led a team of four students to design an automated street cleaner robot for sustainable engineering
- **CAD:** Modeled robot design in Fusion within 36 hours and presented to Judges from industry and academia
- Awarded 2nd place out of 10 teams of engineering students from across Canada

AWARDS AND FELLOWSHIPS

Georgia Tech Mechanical Engineering MsC Award

April 2026

Awarded for outstanding academic and research performance

Georgia Tech Graduate Teaching Fellowship

August 2025

Awarded as one of fifteen graduate students for outstanding teaching and mentorship in 2025. First master's student to be awarded in institute history

National Research Council Metrology RC Award

August 2022

Awarded for contribution above and beyond expectations during the arctic field measurement campaign

Canadian Society of Mechanical Engineering Gold Medal

May 2022

Awarded for having the highest academic standing in the mechanical engineering graduating class